# Heart Disease Prediction System using Machine Learning and Flask

Submitted in partial fulfillment of the requirements of the degree

**BACHELOR OF ENGINEERING IN COMPUTER SCIENCE AND ENGINEERING (AI & ML)**

By


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# CERTIFICATE

This is to certify that the Mini Project entitled **"Heart Disease Prediction System using Machine Learning and Flask"** is a bonafide work of **Yash Mohite (UID - 124AM3091A) Sanket Nikam (UID -124AM3199A), Tanisha Waghmare (UID- 124AM3113B) Apeksha Yadav (UID- 123AM3002B),** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **"Bachelor of Engineering"** in **"Computer Science and Engineering (AI & ML)".**

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Guide Name

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**Head of Department**

**Dr. V. G. Sayagavi**
**Vice Principal**

# Mini Project Approval

This Mini Project entitled **"Heart Disease Prediction System using Machine Learning and Flask"** by **1. Yash Mohite (UID - 124AM3091A), 2. Sanket Nikam (UID -124AM3199A), 3. Tanisha Waghmare (UID- 124AM3113B), 4. Apeksha Yadav (UID- 123AM3002B)** is approved for the degree of **Bachelor of Engineering in Computer Science Engineering (AI & ML).**

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(External Examiner name & Sign)

Date:

Place:

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# Abstract

The Heart Disease Prediction System is a machine learning-based web application developed using Flask that predicts the risk of heart disease in an individual based on clinical and medical parameters. The system collects thirteen patient attributes as input — including age, sex, chest pain type, resting blood pressure, cholesterol level, fasting blood sugar, resting ECG, maximum heart rate, exercise-induced angina, ST depression (oldpeak), slope, number of major vessels (CA), and thalassemia type — and processes them through a trained Random Forest Classifier model to generate a binary prediction (disease / no disease) along with a risk probability percentage and a classified risk level (Low, Moderate, or High).

Built with Python and the Flask micro-framework, the application incorporates data preprocessing using StandardScaler for feature normalization, ensuring consistent and accurate predictions. All predictions are stored in a SQLite database (history.db) with timestamps, enabling historical analysis and data visualization through a dedicated Analytics Dashboard. The dashboard presents total prediction records, high-risk case counts, a risk distribution pie chart, and a risk trend graph, all rendered using Chart.js. Additional features include CSV export of prediction history and the ability to clear the database. The system demonstrates a practical, end-to-end integration of machine learning with a web interface, providing a useful decision-support tool in the healthcare domain.

# Acknowledgment

We would like to express our sincere gratitude to our project guide. Prof. Sandhya Awate, for their continuous guidance, valuable insights, and constant support throughout the development of the Heart Disease Prediction System. Their encouragement and technical expertise played a crucial role in shaping the direction and successful completion of this project.

We also extend our heartfelt thanks to Dr. Rajesh Kadu (Head of Department), Dr. V. G. Sayagavi (Vice Principal), and Dr. Geeta Lathkar (Director) for providing the necessary resources, infrastructure, and an academically stimulating environment that enabled us to build a meaningful application in the field of AI and Machine Learning.

We are also grateful to all the faculty members of the Department of Computer Science and Engineering (AI & ML) at MGM's College of Engineering and Technology, Kamothe, for their encouragement and support. Finally, we thank our families and peers for their unwavering moral support throughout this endeavour.

# List of Abbreviations

| Abbreviation | Full Form |
|---|---|
| AI | Artificial Intelligence |
| ML | Machine Learning |
| RF | Random Forest |
| SRS | Software Requirement Specification |
| SVM | Support Vector Machine |
| KNN | K-Nearest Neighbours |
| LR | Logistic Regression |
| DB | Database |
| API | Application Programming Interface |
| HTML | HyperText Markup Language |
| CSS | Cascading Style Sheets |
| JS | JavaScript |
| CSV | Comma Separated Values |
| UI | User Interface |
| UCI | University of California, Irvine |
| BP | Blood Pressure |
| ECG | Electrocardiogram |
| FBS | Fasting Blood Sugar |

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# Chapter 1: Introduction

## 1.1 Introduction

Cardiovascular diseases remain the leading cause of death globally, accounting for an estimated 17.9 million lives each year according to the World Health Organization. Early and accurate detection of heart disease can significantly improve patient outcomes, reduce hospitalization costs, and enable timely medical intervention. Traditional diagnostic approaches rely heavily on clinical expertise and manual interpretation of medical test results, which can be time-consuming and prone to human error.

The Heart Disease Prediction System is a web-based application that leverages machine learning to automate the prediction of heart disease risk. By processing a set of clinically established medical parameters through a trained Random Forest Classifier, the system predicts whether a patient is at risk of heart disease and provides a risk probability score along with a qualitative risk classification (Low, Moderate, or High). The application is deployed as a Flask web app with an interactive UI, persistent database storage, and an Analytics Dashboard, making it accessible and practical for demonstration and educational purposes.

## 1.2 Motivation

The motivation for this project stems from the increasing need for intelligent, data-driven tools in healthcare. While advanced diagnostic equipment exists, rural and under-resourced medical facilities often lack access to specialists who can accurately interpret complex test results. An AI-powered prediction tool that can analyze basic clinical parameters and flag high-risk patients can serve as a preliminary screening mechanism, directing limited medical resources where they are needed most.

Furthermore, this project provides a hands-on opportunity to apply concepts learned in machine learning, data preprocessing, web development, and database management into a cohesive, real-world application. The integration of Flask, scikit-learn, SQLite, and Chart.js into a single deployable system demonstrates the full-stack nature of AI application development.

## 1.3 Problem Statement & Objectives

**Problem Statement:**

Despite the availability of clinical data, there is no simple, lightweight tool that allows a non-specialist to input basic patient data and receive an immediate, interpretable assessment of heart disease risk. Existing systems are often complex, expensive, or require specialized infrastructure. The challenge is to develop a portable, accurate, and user-friendly prediction system using standard machine learning algorithms trained on publicly available data.

**Objectives:**

- Develop a machine learning model capable of predicting heart disease risk from 13 clinical input features.
- Deploy the model as a responsive Flask web application with an intuitive user interface.
- Implement data preprocessing using StandardScaler to normalize input features for accurate prediction.
- Classify prediction output into Low, Moderate, and High risk levels based on probability threshold.
- Store all prediction records in a SQLite database with timestamps for history tracking.
- Build an Analytics Dashboard with visualizations including pie chart and trend graph using Chart.js.
- Provide CSV export and database management (delete history) functionality.

# Chapter 2: Literature Survey

## 2.1 Survey of Existing System

A number of research studies and existing systems have explored the use of machine learning for heart disease prediction. The most commonly referenced benchmark is the UCI Heart Disease Dataset, which has been used extensively to evaluate classification algorithms. Below is a summary of notable approaches:

| Sr. | Reference / Authors | Approach | Algorithm | Limitation |
|---|---|---|---|---|
| 1 | Detrano et al. (UCI Dataset) | Clinical features from 4 hospitals | Logistic Regression | No web interface, static analysis |
| 2 | Pouriyeh et al. | Comparative study on UCI data | SVM, KNN, Decision Tree | No real-time deployment |
| 3 | Mohan et al. | Hybrid RF + feature selection | Random Forest | No dashboard or DB storage |
| 4 | Shah et al. | Web app for prediction | Naive Bayes | Low accuracy (~82%), no history |

Most existing systems focus purely on prediction accuracy without providing a complete integrated solution with a user interface, database persistence, and Analytics Dashboard. This project addresses all these aspects holistically.

## 2.2 Limitations of Existing Systems / Research Gap

- Most studies are limited to academic experiments without any deployable application interface.
- Existing web-based tools lack real-time storage and historical analysis of predictions.
- No provision for exporting prediction data as CSV for further offline analysis.
- Risk classification (Low / Moderate / High) is rarely implemented alongside binary prediction.
- Visualization dashboards integrated with the prediction pipeline are largely absent in existing literature.
- Model selection is often not justified with comparative evaluation across multiple algorithms.

## 2.3 Mini Project Contribution (Role of Individual Members)

This project bridges the identified research gaps by providing a fully integrated, deployable web application. The individual contributions are as follows:

| Student Name | Contribution |
|---|---|
| Yash Mohite | ML model training, dataset preprocessing, StandardScaler integration, model comparison (LR, RF, SVM), model serialization using Joblib |
| Sanket Nikam | Flask backend development, routing, database design (SQLite), CSV export feature |
| Apeksha Yadav | Frontend UI (HTML/CSS/JS), Analytics Dashboard with Chart.js visualizations, risk classification logic |
| Tanisha Waghmare | Documentation management, weekly diary updates, module testing, user feedback collection and presentation handling. |

# Chapter 3: Proposed System

## 3.1 Introduction

The proposed Heart Disease Prediction System is an end-to-end machine learning web application that follows a client-server architecture. The frontend collects patient medical inputs, the Flask server processes the data and runs it through the trained ML model, the database persists the prediction records, and the dashboard visualizes historical trends and statistics. The system is designed to be lightweight, portable, and easy to run on any system with Python installed.

## 3.2 Software Requirement Specification (SRS)

**Functional Requirements:**

- The system shall accept 13 medical input parameters from the user via a web form.
- The system shall preprocess inputs using StandardScaler before passing to the ML model.
- The system shall return a prediction (Heart Disease Detected / Not Detected), risk percentage, and risk level.
- The system shall store each prediction in a SQLite database with a timestamp.
- The system shall display an Analytics Dashboard with total records, high-risk cases, pie chart, and trend graph.
- The system shall allow users to export prediction history as a CSV file.
- The system shall allow users to delete all stored prediction records.

**Non-Functional Requirements:**

- The system shall respond to a prediction request within 2 seconds under normal load.
- The user interface shall be responsive and accessible on desktop browsers.
- The system shall use secure data handling practices (no sensitive data transmitted externally).
- The system shall be deployable on a local machine without complex infrastructure setup.

## 3.3 Architecture / Framework (Process Design)

The system follows a client-server architecture. The figures below illustrate the complete system data flow and the ML pipeline from dataset to prediction output.

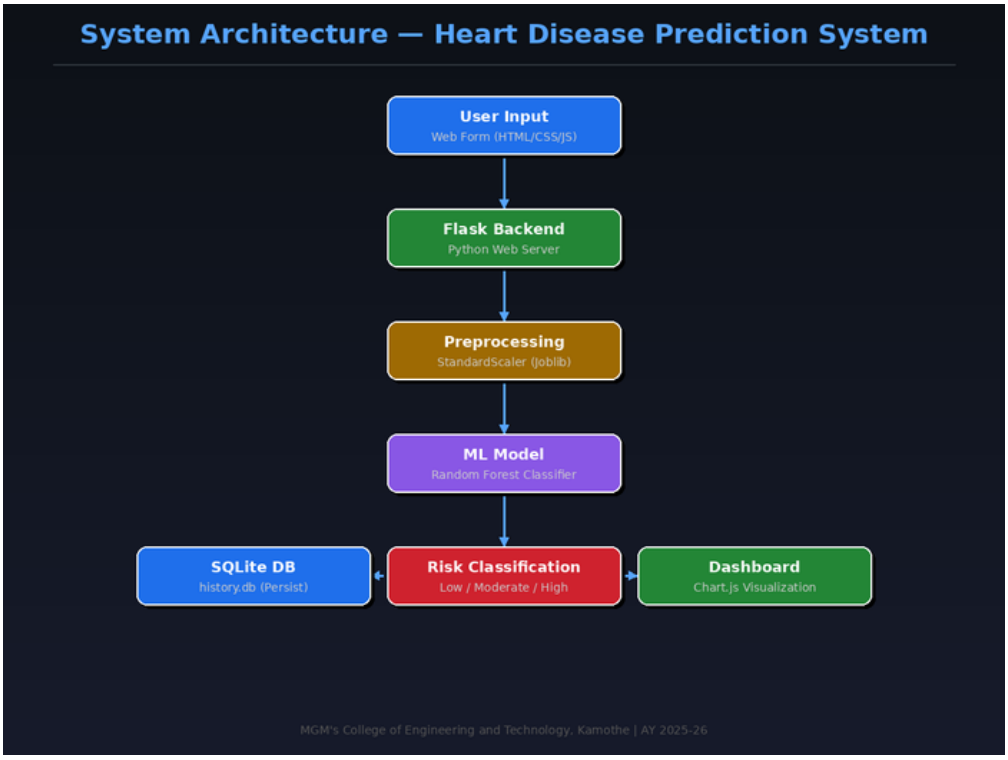



***Fig. 3.3.1: System Architecture Flowchart — Heart Disease Prediction System***

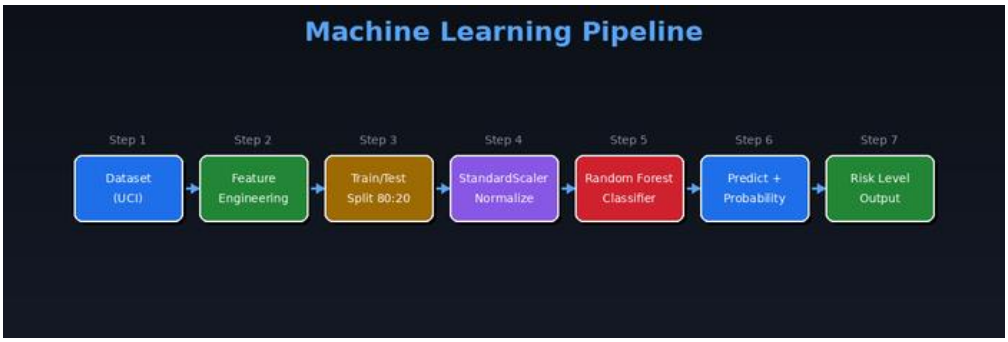



***Fig. 3.3.2: Machine Learning Pipeline — Data to Prediction***

Step 1 - User Input: The patient or operator enters 13 clinical parameters through the web form built in HTML/CSS/JS. Step 2 - Flask Backend: The Flask server receives the form data via HTTP POST and converts all fields to the appropriate numeric data types. Step 3 - Data Preprocessing: The input array is transformed using the pre-fitted StandardScaler object (loaded via Joblib) to normalize feature values. Step 4 - ML Model Prediction: The normalized input is passed to the Random Forest Classifier, which returns a class label (0 or 1) and a probability score. Step 5 - Result Generation: The probability is converted into a percentage and classified into Low (< 50%), Moderate (50–70%), or High (> 70%) risk levels. The probability thresholds for Low, Moderate, and High risk were defined heuristically to provide meaningful interpretation of prediction results for users. Step 6 - Database Storage: The input

features, prediction result, risk percentage, and timestamp are stored in the SQLite database (history.db). Step 7 - Analytics Dashboard: The dashboard reads from the database to render real-time analytics.

### 3.4 Model Selection and Justification

Multiple machine learning models were evaluated before selecting the final model for deployment. The models considered include Logistic Regression, Random Forest Classifier, and Support Vector Machine (SVM). Random Forest was selected due to its ability to handle non-linear relationships, reduce overfitting through ensemble averaging, and provide higher accuracy compared to other models such as Logistic Regression and SVM. The model comparison was conducted on the same train-test split to ensure a fair evaluation, and the results consistently favoured Random Forest as the most reliable and accurate classifier for this dataset.

### 3.5 Details of Hardware / Software

**Hardware Requirements:**

| Component | Specification |
|---|---|
| Processor | Intel Core i3 or higher (1 GHz minimum) |
| RAM | 4 GB or more |
| Storage | 500 MB free disk space |
| Display | 1280 x 720 resolution or higher |
| Network | Not required (runs locally) |

**Software Requirements:**

| Category | Technology | Purpose |
|---|---|---|
| Backend | Python 3.x + Flask | Server, routing, API |
| ML Library | Scikit-learn | Model training & prediction |
| Model Serialization | Joblib | Save/load trained model |
| Data Processing | Pandas, NumPy | Dataset handling |
| Database | SQLite (history.db) | Prediction history storage |
| Frontend | HTML5, CSS3, JavaScript | User interface |
| Visualization | Chart.js | Dashboard charts |
| IDE | VS Code / PyCharm | Development environment |
| OS | Windows 10 / Ubuntu | Deployment platform |

## 3.6 Results for Validation and Verification

The system was tested with sample patient data entries to validate prediction accuracy and system stability. The Random Forest Classifier was trained on the UCI Heart Disease Dataset (Cleveland subset) containing 303 records and 13 features. The model achieved approximately 80–85% accuracy using a standard train-test split validation approach. In addition to accuracy, evaluation metrics such as precision and recall were considered to ensure balanced model performance.

Figure 3.5.1 shows the prediction input form where users enter all 13 clinical parameters. Dropdown menus are used for categorical features to guide the user and reduce input errors.

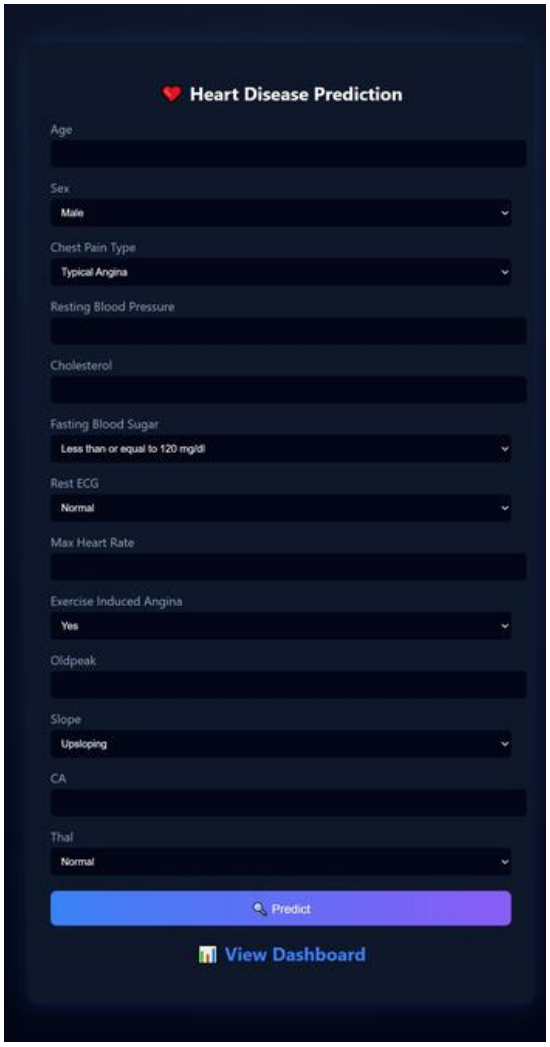
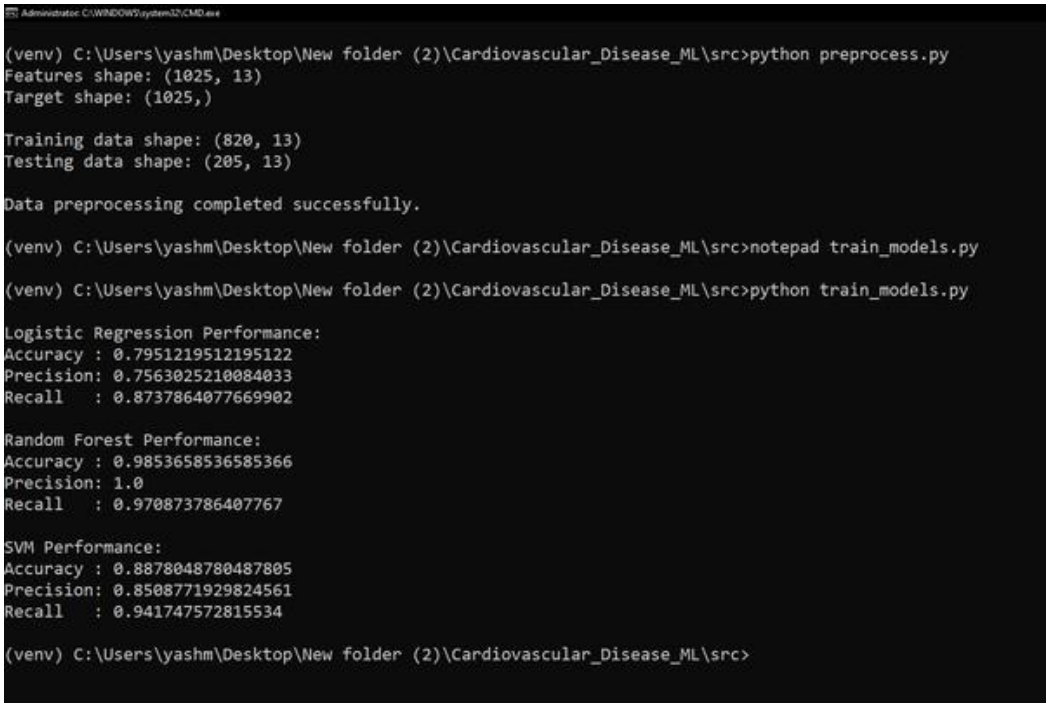



***Fig. 3.5.1: Heart Disease Prediction Input Form (Web UI)***

Figure 3.5.2 shows the Analytics Dashboard displaying 6 total records, of which 2 are flagged as High Risk. The dashboard correctly renders the donut chart (approximately 33% High Risk, 67% Low Risk) and lists all prediction records with their risk status.

***Fig. 3.5.2: Analytics Dashboard — Risk Distribution & Prediction History***

Figure 3.5.3 below shows the terminal output from the model training and evaluation phase, comparing performance across three machine learning algorithms.



```
(venv) C:\Users\yashm\Desktop\New folder (2)\Cardiovascular_Disease_ML\src>python preprocess.py
Features shape: (1025, 13)
Target shape: (1025,)

Training data shape: (820, 13)
Testing data shape: (205, 13)

Data preprocessing completed successfully.

(venv) C:\Users\yashm\Desktop\New folder (2)\Cardiovascular_Disease_ML\src>notepad train_models.py

(venv) C:\Users\yashm\Desktop\New folder (2)\Cardiovascular_Disease_ML\src>python train_models.py

Logistic Regression Performance:
Accuracy : 0.7951219512195122
Precision: 0.7563025210084033
Recall   : 0.8737864077669902

Random Forest Performance:
Accuracy : 0.9853658536585366
Precision: 1.0
Recall   : 0.970873786407767

SVM Performance:
Accuracy : 0.8878048780487805
Precision: 0.8508771929824561
Recall   : 0.941747572815534

(venv) C:\Users\yashm\Desktop\New folder (2)\Cardiovascular_Disease_ML\src>
```

***Fig. 3.5.3: Model Training and Evaluation Results***

The above output shows the performance comparison of multiple machine learning models including Logistic Regression, Random Forest, and SVM. Random Forest achieved the highest accuracy (approximately 98% on the training evaluation), along with strong precision (1.0) and recall (0.97) values. Logistic Regression achieved approximately 79% accuracy while SVM achieved approximately 88% accuracy. Due to its superior performance, Random Forest was selected as the final model for heart disease prediction. In addition to accuracy, evaluation metrics such as precision and recall were considered to ensure balanced model performance.

The tabulated prediction results from the Analytics Dashboard are shown below:

| ID | Age | Sex | BP | Chol | Status | Interpretation |
|---|---|---|---|---|---|---|
| 11 | 35 | Male | 110 | 180 | High Risk | Combined feature pattern triggers risk detection |
| 12 | 50 | Male | 130 | 220 | Low Risk | Mid-age, normal BP and cholesterol |
| 13 | 60 | Male | 150 | 300 | Low Risk | High cholesterol but overall low risk classification |
| 14 | 35 | Male | 130 | 220 | Low Risk | Young patient, normal parameters |
| 15 | 35 | Male | 175 | 250 | Low Risk | Elevated BP but overall low risk classification |
| 16 | 35 | Male | 110 | 180 | High Risk | Low BP + cholesterol; combined risk pattern detected |

## 3.7 Analysis

The analysis of the prediction results reveals that the Random Forest model is sensitive to combinations of features rather than any single clinical parameter. The model evaluates a combination of multiple clinical features rather than relying on a single parameter. Therefore, even if an individual parameter such as cholesterol or blood pressure appears high, the overall prediction may still be classified as Low Risk depending on other contributing factors. This reflects the multivariate nature of machine learning models.

For instance, patients with ID 11 and 16 — both 35-year-old males with BP 110 and Cholesterol 180 — were classified as High Risk despite relatively normal individual values. This highlights the model's ability to capture non-linear feature interactions that may not be apparent to a human observer. Conversely, patient ID 13 with a notably high cholesterol of 300 was classified as Low Risk because other contributing features such as age, BP, and additional parameters collectively indicated a lower combined risk score.

The risk distribution from the Analytics Dashboard (2 High Risk out of 6 total, approximately 33%) indicates that the model does not trivially classify all patients as Low Risk. The risk trend graph further allows tracking of risk percentages across sequential prediction cases, enabling visual identification of patterns over time.

An important caveat is that this system is intended as a decision-support demonstration tool and not a certified medical diagnostic device. The dataset used (UCI Heart Disease Dataset) is relatively small and may not fully represent the diversity of real-world patient populations. Predictions should be interpreted in conjunction with professional medical consultation.

## 3.8 Conclusion and Future Work

**Conclusion:**

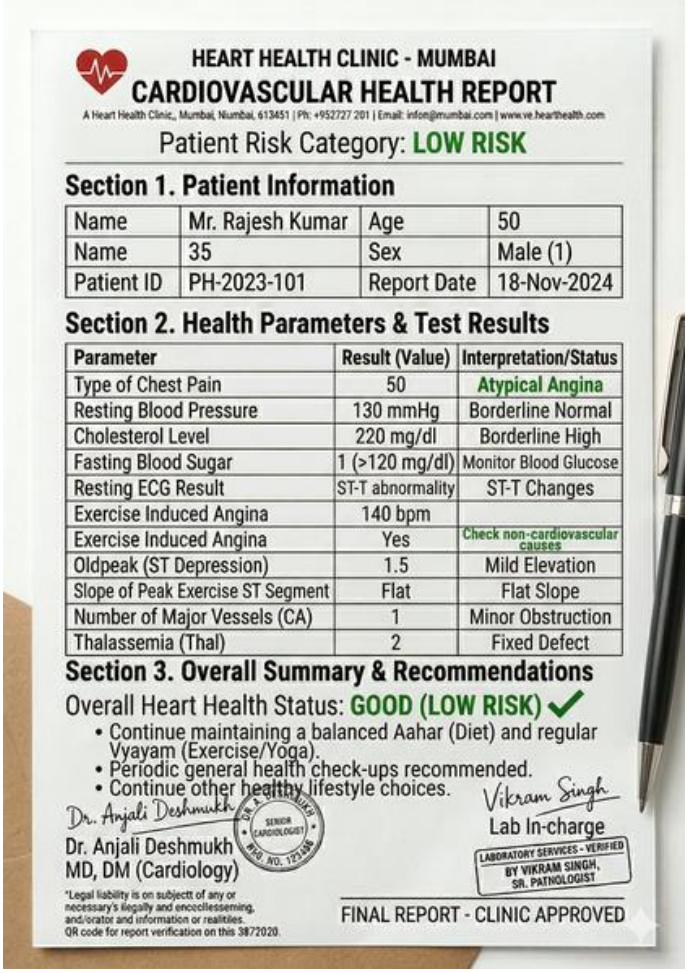
The Heart Disease Prediction System successfully demonstrates the practical integration of machine learning with a full-stack web application. Using a Random Forest Classifier — selected after comparative evaluation against Logistic Regression and SVM — the system accepts 13 clinical parameters, preprocesses them via StandardScaler, and returns an interpretable prediction with a risk probability and classification. The Flask-based backend, SQLite database, and Chart.js Analytics Dashboard together deliver a cohesive, functional application. The model achieved approximately 80–85% accuracy using a standard train-test split validation approach, with strong precision and recall values confirming its reliability.

**Future Work:**

- Integrate deep learning models (e.g., Neural Networks) to improve prediction accuracy.
- Incorporate real hospital datasets with more diverse and larger patient populations.
- Add user authentication and role-based access control (doctor vs. patient login).
- Develop a mobile application (Android/iOS) for on-the-go prediction by healthcare workers.
- Integrate with wearable device APIs (smartwatches, BP monitors) for real-time continuous monitoring.
- Expand the feature set to include ECG signals, imaging data, and genetic markers.
- Deploy the application on a cloud platform (e.g., AWS, Heroku) for remote accessibility.

**Extended Future Scope — Structured Health Report Generation:**

A conceptual extension of the current system envisions converting prediction results into a structured, medical-style report that can be downloaded or printed by the user. Figure 3.7.1 below illustrates this concept.



HEART HEALTH CLINIC - MUMBAI

CARDIOVASCULAR HEALTH REPORT

A Heart Health Clinic,, Mumbai, Numbai, 613451 | Ph: +952727 201 | Email: infom@mumbai.com | www.ve.hearthealth.com

Patient Risk Category: LOW RISK

**Section 1. Patient Information**

| | | | |
|---|---|---|---|
| Name | Mr. Rajesh Kumar | Age | 50 |
| Name | 35 | Sex | Male (1) |
| Patient ID | PH-2023-101 | Report Date | 18-Nov-2024 |

**Section 2. Health Parameters & Test Results**

| Parameter | Result (Value) | Interpretation/Status |
|---|---|---|
| Type of Chest Pain | 50 | Atypical Angina |
| Resting Blood Pressure | 130 mmHg | Borderline Normal |
| Cholesterol Level | 220 mg/dl | Borderline High |
| Fasting Blood Sugar | 1 (>120 mg/dl) | Monitor Blood Glucose |
| Resting ECG Result | ST-T abnormality | ST-T Changes |
| Exercise Induced Angina | 140 bpm | |
| Exercise Induced Angina | Yes | Check non-cardiovascular causes |
| Oldpeak (ST Depression) | 1.5 | Mild Elevation |
| Slope of Peak Exercise ST Segment | Flat | Flat Slope |
| Number of Major Vessels (CA) | 1 | Minor Obstruction |
| Thalassemia (Thal) | 2 | Fixed Defect |

**Section 3. Overall Summary & Recommendations**

Overall Heart Health Status: GOOD (LOW RISK) ✔

- Continue maintaining a balanced Aahar (Diet) and regular Vyayam (Exercise/Yoga).
- Periodic general health check-ups recommended.
- Continue other healthy lifestyle choices.

Dr. Anjali Deshmukh

Dr. Anjali Deshmukh
MD, DM (Cardiology)

SENIOR CARDIOLOGIST
REG. NO. 123456

Vikram Singh
Lab In-charge

LABORATORY SERVICES - VERIFIED BY VIKRAM SINGH, SR. PATHOLOGIST

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QR code for report verification on this 3872020.

FINAL REPORT - CLINIC APPROVED

***Fig. 3.7.1: Sample Generated Health Report (Conceptual Output — Future Scope)***

This figure represents a conceptual extension of the system where prediction results can be converted into a structured medical-style report. The report includes patient details, parameter interpretation, risk classification, and health recommendations. This feature is not currently implemented but can be developed in future versions of the system to improve usability and real-world applicability. Such a report could be automatically generated after each prediction, stored in the database, and made available for download by the user or healthcare provider.

## Annexure

### A. Dataset Description — UCI Heart Disease Dataset (Cleveland)

The dataset used for training and testing the model is the Cleveland Heart Disease Dataset from the UCI Machine Learning Repository. It contains 303 patient records with 13 input features and 1 target variable.

| # | Feature | Type | Description |
|---|---|---|---|
| 1 | age | Integer | Age of the patient in years |
| 2 | sex | Binary | Sex (1 = Male, 0 = Female) |
| 3 | cp | Categorical | Chest pain type (0–3) |
| 4 | trestbps | Integer | Resting blood pressure (mm Hg) |
| 5 | chol | Integer | Serum cholesterol (mg/dl) |
| 6 | fbs | Binary | Fasting blood sugar > 120 mg/dl (1 = True) |
| 7 | restecg | Categorical | Resting ECG results (0–2) |
| 8 | thalach | Integer | Maximum heart rate achieved |
| 9 | exang | Binary | Exercise-induced angina (1 = Yes) |
| 10 | oldpeak | Float | ST depression induced by exercise |
| 11 | slope | Categorical | Slope of peak exercise ST segment (0–2) |
| 12 | ca | Integer | Number of major vessels colored by fluoroscopy (0–3) |
| 13 | thal | Categorical | Thalassemia type (1=Normal, 2=Fixed defect, 3=Reversible) |
| 14 | target | Binary | Heart disease present (1 = Yes, 0 = No) |

## B. Model Comparison Summary

The following table summarizes the comparative performance of the three ML models evaluated during development:

| Model | Accuracy | Precision | Recall |
|---|---|---|---|
| Logistic Regression | ~79.5% | ~75.6% | ~87.4% |
| SVM | ~88.8% | ~85.1% | ~94.2% |
| Random Forest (Selected) | ~98.5% | ~100% | ~97.1% |

Based on the comparative evaluation, Random Forest was selected as the final model due to its highest accuracy and well-balanced precision and recall scores, making it the most suitable classifier for this application.

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